COLLEGE OF COMPUTER STUDIES

CAPSTONE MANUAL

PROGRAMS

- Bachelor of Science in Information Technology (B.S.I.T.)

VISION

A God- loving educational community with passion for truth and compassion for humanity.

MISSION

We commit ourselves to the:

1. Total formation of the person
2. Promotion of truth
3. Transformation of values for the service of humanity

GOALS

We aim to provide transformative education that is responsive to the development of Wisdom, Social Responsibility, and Christian Witness in accordance with the Gospel Values to become a productive member of the society.

OBJECTIVES

A. Wisdom

Cognitive - Possess knowledge and skills for social communication.
Affective - Demonstrate sensitive awareness of one's role in social building
Psychomotor - Positively engage oneself in relevant social issues

B. Social Responsibility

Cognitive - Inculcate awareness of Filipino Christian values
Affective - Show appreciation of our Christian dignity as stewards of God's creation
Psychomotor - Promote moral commitment to ecological balance

C. Christian Witness

Cognitive - Acquire understanding scriptural teachings
Affective - Manifest love for the scripture
Psychomotor - Practice scriptural truths in every aspect of life

Institutional Graduate Outcomes

Competent • Disciplined • Life-long Learner • Steward

Program Graduate Attributes	**Program Graduate Outcomes Code**	**Program Graduate Outcomes**
Knowledge for Solving Computing Problems	IT01	Apply knowledge of computing, science, and mathematics appropriate to the discipline
	IT02	Understand best practices and standards and their applications
Problem Analysis	IT03	Analyze complex problems, and identify and define the computing requirements appropriate to its solution
	IT04	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems
Design/Development of Solutions	IT05	Design, implement, and evaluate computer-based systems, processes, components or programs to meet desired needs and requirements under various constraints
	IT06	Integrate IT-based solutions into the user environment effectively
Modern Tool Usage	IT07	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession
Individual and Team Work	IT08	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal
	IT09	Assist in the creation of an effective IT project plan
Communication	IT10	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.
Computing Professionalism and Social Responsibility	IT11	Analyze the local and global impact of computing information technology on individuals, organizations, and society
	IT12	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology
Life-Long Learning	IT13	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development

TABLE OF CONTENTS

1.0 Introduction

Based on the guidelines for implementation of **CMO 25 s. 2015**, a Capstone Project is required for BSIT as a terminal project requirement that would not only demonstrate a student's comprehensive knowledge of the area of study and research methods used but also allow them to apply the concepts and methods to a specific problem in there are of specialization.

BS Information Technology students must complete a capstone project such as software/system development with emphasis on the IT infrastructure, a Multimedia Systems development, or an IT Management project.

It is expressly understood that Capstone Projects need not require surveys, statistics, and descriptive methods, unless appropriate.

A **Capstone Project** is an undertaking appropriate to a professional field. It should significantly address an existing problem or need. An Information Technology Capstone Project focuses on the infrastructure, application, or processes involved in implementing a Computing solution to a problem.[1]

2.0 Scope of Capstone Projects

The Capstone Project should integrate the different courses, knowledge, and competencies learned in the curriculum. Students are encouraged to produce innovative results, generate new knowledge or theories or explore new frontiers of knowledge or application ideas.

For Information Technology Capstone Projects, recommended infrastructure and its implications on other systems should be clearly specified in the final report with the introduction of the project.

The capstone project adviser should determine the appropriate complexity level of the specific problem being addressed and the proposed solution, considering the duration of the project, the composition of the team, and the resources available.[2]

3.0 Suggested Areas for Capstone Projects

Following is a list of some suggested areas under Information Technology.

3.1. Software Development

- Software Customization
- Information Systems Development for an actual client (with pilot testing)
- Web Application Development (with at least alpha testing on live servers)
- Mobile Computing Systems

3.2. Multimedia Systems

- Game Development

[1] Section 8.4 CMO 25 s 2015, p. 15
[2] Section 3 Annex A CMO 25 s. 2015, p. 50

- e-Learning Systems
- Interactive Systems
- Information Kiosks

3.3. Network Design and Implementation and Server Farm Configuration and Management
3.4. IT Management

- IT Strategic Plan for sufficiently complex enterprises
- IT Security Analysis, Planning and Implementation[3]

4.0 Project Duration

Students should be given ample time to finish their project. One (1) to three (3) terms or semesters should be prescribed in the curriculum for BS Information Technology students to complete their Capstone Projects.

The maximum number of units that may be required for Capstone Projects is nine (9) units.

Grading systems and possible honoraria rates for capstone project are left to the discretion of the HEI, provided that such policies are not grossly disadvantageous to the students, and provided further that such policies are documented and approved by the proper HEI authorities.[4]

The Capstone Project shall be completed within the prescribed period of time in accordance of the curriculum and with the following phases:

Pre-Proposal Stage

- ✓ Course Enrolment
- ✓ Capstone Project Orientation
- ✓ Short Listing of Possible Capstone Projects

Proposal Stage

- ✓ Title Proposal and Critiquing (with Patentability Check if possible)
- ✓ Writing of Chapters 1, 2, 3 and 4 (planning and design sections only of chapter 4)
- ✓ Proposal Manuscript Submission
- ✓ Oral Defense
- ✓ Proposal Manuscript Revisions

Final Defense Stage

- ✓ Analysis
- ✓ Design
- ✓ Development
- ✓ Testing
- ✓ Capstone Project Manuscript Submission
- ✓ Final Defense Proper
- ✓ Capstone Project Manuscript Revisions

[3] Section 4 Annex A CMO 25 s. 2015, p.52
[4] Section 5 Annex A CMO 25 s. 2015, p.53

- ✓ Public Presentation (see Chapter 7.0)
- ✓ Submission of Final Requirements

Patent Process (optional)

- ✓ Patent Drafting
- ✓ Patent Application (if possible)
- ✓ Technology Transfer

5.0 Composition of Project Groups

Students should preferably work in teams of two (2) to four (5) members depending on the complexity of the project. The adviser should be able to determine whether the team can complete the project on time.

Multidisciplinary teams are also encouraged, provided that team members prepare separate documentations per program.[5]

5.1. The following are the roles that the proponents/researchers should play:

Project Manager (PM) – The person with authority to manage a Capstone Project. This includes leading the planning and the development of all Capstone Project deliverables. The project manager is responsible for the budget, work plan, and all Project Management Procedures (scope management, issues management, risk management, etc.). He is responsible for the success of the entire activity.

Systems Analyst / Database Designer (SA/DD) – The person who checks that all parts of the system are coordinated. The person who makes sure that the database design is complete and robust. He coordinated well with the PM.

Network Designer / UI Designer (ND/UID) – The person who masters the system's the network design and prepares the user-interface design (forms, screens shots, storyboard). He coordinated well with the SA/DD.

Software Engineer / Programmer (SE/P) – The person who design, write, and test computer programs. He coordinates well with the ND/ID. May be two (2) in the project group (Co-Programmer).

QA Tester / Technical Writer (QA/TW) – A person who ensures the quality of the software product and help find and eliminate any bugs. He determines the functionality of every aspect of a particular application. A person who finalizes the Capstone Project study document, both the system and the Capstone Project manuscript. He coordinates well with the SE/P.

5.2. Duties and Responsibilities of the Proponents/Researchers

a. Keep informed of the Capstone Project Guidelines and Policies.
b. Keep informed of the schedule of Capstone Project activities, required deliverables and deadlines posted by Adviser and Dean.

[5] Section 6 Annex A CMO 25 s. 2015, p.53

c. Submit on time all deliverables specified in this document as well as those to be specified by the Adviser and Dean.
d. Submit on time all requirements identified by the Capstone Project Oral Defense Panel during the Oral Defense.
e. Submit on time the requirements identified by the Adviser throughout the duration of the Capstone Project.
f. Schedule regular meetings (at least once a month) with the Adviser throughout the duration of the Capstone Project. The meetings serve as a venue for the proponents/researchers to report the progress of their work, as well as raise any issues or concerns.
g. Schedule regular meetings (at least once in a semester) with the Dean throughout the duration of the Capstone Project.

5.3. Policy on Regrouping

Regrouping is allowed if less than three (3) members of the group remain from Capstone 1 to Capstone 2. Should this happen, the group may be disbanded and members of these affected groups may join in other groups for as long as the maximum number for each group is followed. However, if the remaining member(s) decide(s) to continue with his/their Capstone Project, regrouping may not apply but with the consent of the Adviser and the Dean. Revision of the scope may then be an option. The title to be pursued will then be decided among the team members and the Dean.

6.0 Adviser/Panel Composition

6.1. Panel Composition

The project is prepared under the guidance of an adviser and presented and accepted by a Panel composed of 1 Chairman, 2 members, and may include content experts and recorder as assigned if necessary.

6.2. Duties and Responsibilities of the Panel

Their duties and responsibilities include the following, but not limited to:

Chairman

a. Brief the proponents/researchers about the Title Proposal or Oral Defense program during the actual Title Proposal or Oral Defense respectively.
b. Issue the verdict. The verdict is unanimous decision among the three members of the Capstone Project Title Proposal or Oral Defense panel. Once issued, it is final and irrevocable.
c. Nominate a Capstone Project for the Capstone Project Award. Guidelines for the Capstone Project Award will be provided separately.

Panel Members / Context Expert

a. Validate the endorsement of the Adviser.
b. Evaluate the deliverables.
c. Recommend a verdict.
d. Listen and consider the request of the Adviser and/or the proponents/researchers.
d. Nominate a Capstone Project for the Capstone Project Award. Guidelines for the Capstone Project Award will be provided separately.

6.3. Duties and Responsibilities as the Adviser

a. Ensures that the study proposed by the students conforms to the standard of the College and has immediate or potential impact on the research thrust of the school.
b. Guides the Capstone Project students in the following tasks while in the proposal stage:
   b.1. Defining the research problems/objectives in clear specific terms
   b.2. Building a working bibliography for the research
   b.3. Identifying variables and formulating hypothesis if any
   b.4. Determining research design, population to be studied, research environment, instruments to be used and the data collection procedures.
c. Meets the teams regularly (at least once a month, NOTE: the team must seek proper appointment) to answer questions and help resolve impasses and conflicts.
d. Points out errors in the development work, in the analysis, or in the documentation. The adviser must remind the proponents/researchers to do their work properly.
e. Reviews thoroughly all deliverables at every stage of the Capstone Project, to ensure that they meet the department's standards. The adviser may also require his/her proponents/researchers to submit progress reports regularly.
f. Recommends the proponents/researchers for Title Proposal and Oral Defense. The Adviser should not sign the Title Proposal Notice and the Oral Defense notice if he/she believes that the proponents/researchers are not yet ready for Title Proposal and Oral Defense, respectively. Thus, if the proponents/researchers fail in the Title Proposal or Oral Defense, it is partially the Adviser's fault.
g. Clarifies points during the Title Proposal and Oral Defense.
h. Ensures that all required revisions are incorporated into the appropriate documents and/or software.
i. Keeps informed of the schedule of Capstone Project activities, required deliverables and deadlines.
j. Recommends to the Title Proposal and Oral Defense panel the nomination of his/her Capstone Project for an award.

6.4. Adviser/Panel Qualifications

The adviser must have completed a computing project successfully beyond the bachelor's degree project. As much as possible, the adviser should be a full-time faculty member of the HEI. Otherwise a full-time co-adviser is required.

Advisers and Panel Members should have a degree in Computing or allied programs, or must be domain experts in the area of study. At least one of the panel members must have a master's degree in Computing (preferably in the same field as the project) or allied program. For IT, at least one of the panel members should preferably have industry experience.

The adviser must be able to guide the students throughout the whole project life cycle, including the capstone project defense and possible project deployment.

Faculty advisers should preferably handle at most five projects at one time, and in no case should exceed ten (10) projects. Panel members should preferably be limited to at most ten (10) projects and in no case should exceed twenty (20) projects in one semester, counting all projects in all HEIs.

In case of the participation of an external client, then the organization for the project is intended should be represented as much as possible.

7.0 Presentation of the Capstone Projects

Capstone project must be presented in a public forum. This forum may be an international, national, regional, or school-based conference, meeting, or seminar that is announced and open to interested parties. This may be separate from the presentation before the Panel mentioned in Chapter VI. A school-based colloquium organized for this purpose would suffice to satisfy this requirement. Presentation in a public forum, such as National Conference on IT Education (NCITE) of PSITE, is encouraged.

8.0 Grading System

8.1 The Final Grade of each proponent will comprise of the following:

Rubric of Capstone Final Grade	
Average grade of the Panel Members including the Chairman	60%
Adviser of the Capstone Project	30%
Co-Researcher (Peer Grading)	10%
TOTAL	100%

8.2 The rating of each proponent per panel member shall be based on the following rubric for objective evaluation purposes:

A. Capstone 1

Rubric for Panel Member Grading – Capstone 1	
Capstone Manuscript (Group Grade)	40%
Oral Examination (Individual Grade)	20%

B. Capstone 2

Rubric for Panel Member Grading – Capstone 2	
Capstone Manuscript (Group Grade)	10%
Capstone Software (Group Grade)	30%
Oral Examination (Individual Grade)	20%

8.3 Rubric of Capstone Manuscript

Rubric of Capstone Manuscript Grading	
Initial Pages • Table of contents is consistent • Acknowledgement is brief and formal • Abstract is brief but complete	4
Chapter 1 • Introduction is intact and provides clear overview of the entire Capstone Project • Statement of the Problem/ Objectives is SMART • Scope and Limitations of the Capstone Project are clearly defined	10
Chapter 2 • Related literatures are recent and relevant • Anchor provides solid background of the Capstone Project • Auxiliary theories are evident • Sources are appropriately cited and noted • Related studies are relevant and includes global and local scope	8

Chapter 3 • There should be comprehensive discussions on the technologies(hardware/software) involved in the Research / Capstone Project and its related Capstone Projects in the past	8
Chapter 4 • Methodology strictly follows the SDLC (esp. for Software Development) • Methodology includes project management techniques appropriate for the chosen Capstone Project. • Requirements Specification is more or less complete and answers the objectives • Design Tools used are relevant and appropriate which should be based on requirements • Development Plan is concrete and should be consistent with the Design • Testing techniques to be used should assess all aspects of the developed Capstone Project • Implementation Plan should be aligned with the objectives	10
Final Pages • Findings and Conclusions are attuned with the objectives • Recommendations are feasible and practical • Terms in the glossary are defined operationally • Bibliography should be in MLA Format • Appendices are relevant and help support the principal content • Glossary should be arranged alphabetically and defined operationally	3
Appendices • Deliverables compiled are in-tact and complete	2
Manuscript Mechanics • Organization and Fluidity of ideas are apparent • Formatting and layout are consistent • All parts of the manuscript should be grammatically correct	5

8.4 Rubric of Capstone Software

Rubric of Capstone Software	
The output should be consistent with the objectives as defined during the proposal stage	10
All major modules and features of the system's output as defined after the proposal stage are delivered. The credit shall be based on the percentage of delivered items.	10
System design and aesthetics	3
Group Debugging • The team shall display competence in resolving planted bugs.	7

8.5 Rubric of Oral Examination

Rubric of Oral Examination Grading	
Comprehensiveness of the Answer/Ideas	10
Contribution/Support to the Team	7

Delivery / Command of the English Language	3

8.6 Rubric for Adviser

Rubric of Adviser's Grade	
Deliverables	20
Attendance	5
Journal Entries / Attitude / Behavior	5

9.0 Verdicts

There will be three (3) possible verdicts after the Capstone Proposal. The verdict is a unanimous decision among the three members of the Capstone Proposal Defense panel. Once issued, it is final and irrevocable.

9.1 Capstone 1

APPROVED. Minor revisions are necessary but they do not have to be presented in front of and checked by all panelists. 35 – 40 (based on proposal manuscript score)
APPROVED WITH REVISIONS. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. 24 - 34
DISAPPROVED. The Proponents failed to propose a researchable or scholarly Capstone Project. Below 24

9.2 Capstone 2

ACCEPTED WITH REVISIONS. Revisions are necessary but they do not have to be presented in front and checked by all panelists. 31 to 50 (based on lowest score in the group of combining Capstone Software and Oral Examinations Scores)
REORAL DEFENSE. Another Oral Defense session, in which all panelists must be present, is necessary to further clarify the objectives and scope of the capstone project. Student must re-apply for another Oral Defense. 21 to 30 and upon the panel's unanimous decision
NOT ACCEPTED. The proponent failed to achieve the objectives of the research established in the proposal. The panelists' numeric grades are not anymore needed. Below 21

10.0 Documentation Guidelines

10.1 Documentation Outline

Title Page
Adviser's Recommendation Sheet
Dean's Acceptance Sheet
Panel's Approval Sheet
Acknowledgement
Abstract
Table of Contents
List of Tables
List of Figures
List of Notations (optional)
1.0 Introduction

10.2 Documentation Contents

Title Page i (does not appear)

The title of the project should be concise as well as descriptive and comprehensive. It is a distinctive name given to the project describing the work scope in a specific context and indicates the content of the paper.

Generally, project titles begin with a noun or present participle form of an action verb. It is advisable to state project titles in general terms in order to bundle a number of problem areas. Humorous or catchy titles are not appropriate and is discouraged.

Adviser's Recommendation Sheet ii (does not appear)

No final capstone documentation will be accepted if all copies are not duly signed by the capstone adviser.

The Adviser's Recommendation Sheet provides space for the signature of the adviser of the group indicating that he/ she has examined and recommended the capstone for acceptance and approval.

Dean's Acceptance Sheet iii (does not appear)

No final capstone documentation will be accepted if all copies are not duly signed by the dean. The college acceptance sheet provides space for the signatures of the Dean of the college indicating their acceptance of the work.

Panel's Approval Sheet iv (does not appear)

No final capstone documentation will be accepted if all copies are not duly signed by all the defense panelists.

The approval sheet provides space for the signatures of the members of the panel indicating their acceptance of the work.

Acknowledgement v
Abstract vi
From 150 to 200 words of short, direct and complete sentences, the abstract should be informative enough to serve as a substitute for reading the project paper itself. It states the rationale and the objectives of the project. Do not put citations or quotes in this section. Avoid beginning the abstract with *"This paper/document/project/study/project/…."*

Table of Contents vii
Observe the following format:

1.0 Introduction	1
1.1 Project Description	2
1.2 Project Objectives	…
1.2.1 General Objective	…
1.2.2 Specific Objectives	…

…

List of Tables ... (roman numeral)
(in bold characters, font size 14)

Table <chapter#>-<table#> <Table Caption> <page>

Examples:

Table 1-2	Percentage Ratio of Sophomore vs. Seniors	6
Table 3-4	Mortality Rate of THESIS2 vs. School Year	7

List of Figures ... (roman numeral)
(in bold characters, font size 14)

Figure <chapter#>-<figure#> <Figure Caption> <page>

Examples:

Figure 2-2	Systems Development Life Cycle	16
Figure 4-1	Data Flow Diagram	32

List of Notations (optional) ... (roman numeral)

Note that the page numbering for preliminary pages like title page, etc. is based on roman numerals while the page numbering for main body of the document is based on decimal numbers. Thus the first page of Chapter 1 is at 1.

1.0 Introduction

[Introduction Proper]

1.1 Project Context

The proponent should introduce the presentation of the problem that is, what the problem is all about. The proponent should describe the existing and prevailing problem situation. This scope may be global, national or regional and local.

The proponent should give strong justification for selecting such research problem in his/her capacity as a researcher. Being part of the organization or systems and the desire and concerned to improve the systems.

The researcher state a sentence or two that would show the link and relationship of the rationale of the study to the proposed researched problem.

Note: Must be at least two pages of presentation and discussions.

The following guide questions may also be used:
What is the function of the project?
What is good in your project?
What makes your project unique, innovative, and relevant?

The Project Context should also include a discussion of the relevant topics, concepts, and technologies that will be utilized in developing the software.

1.3 Objectives

1.3.1 General Objective/Major Objective

This should contain a single paragraph describing the general objective of the capstone project.

1.3.2 Specific Objectives/Minor Objectives

This should contain a list of the specific work that the proponents expect to address to arrive at the accomplishment of the general objective.

Note: Objectives should be **SMART**.

Specific: The problem should be specifically stated.

Measurable: It is easy to measure by using research instruments, apparatus, or equipment.

Achievable: Solutions to a research problems are achievable feasible.

Realistic: Real results are attained because they are gathered scientifically and not manipulated or maneuvered.

Time-Bound: Time frame is required in every activity because the shorter completion of the activity, the better.

1.4 Scope and Limitations

This should contain the extent of the prototype to be developed and the means by which the proposed system is to be evaluated on its capability of solving the problem.

The limitations are those conditions beyond the control of the proponents that may place restrictions on the conclusions of the study and their application to other situations. (Justify each limitation.)

2.0 Review of Related Literature / Systems

This portion of project proposal contains presentations and discussions of the following two (2) components:

- **Related Theories**
  Outline first, starting off with an anchor theory.

Supporting theories help elaborate the anchor theory.
Endnoting and footnoting is important which follows correct bibliography entry.
Fluidity and Continuity should be observed.

- **Related Projects**
  Overview of the current system/project.
  Inventory of every related and existing project/system.
  Fluidity and continuity should be observed.
  Comparative matrix should be more appropriate.
  Screen shots help make the presentation believable.
  May consider 3 to 6 related studies/projects.

3.0 Technical Background

Overview of the current technologies (hardware/software/network) used in the current system.

Discussions on the current trends and technologies to be used in developing and implementing the proposed system

- Hardware
- Software
- Peopleware
- Network

✓ Fluidly and continuity should be observed.
✓ Technically of the project.
✓ Details of the project
✓ How to project with work.

4.0 Methodology, Results and Discussion

4.1 Methodology

Identifies the formal method that the proponents intend to follow in order to accomplish what have been set in the objectives. The formal methodologies are any of the software engineering systems analysis and design methodologies.

4.2 Environment (only for organization-specific capstone project)

4.2.1 Locale

4.2.2 Population of the Study

4.2.3 Organizational Chart/Profile

4.3 Requirements Specification

4.3.1 Operational Feasibility

- Fishbone Diagram
- Functional Decomposition Diagram

4.3.2 Technical Feasibility

- Compatibility Checking (hardware/software and other technologies)
- Relevance of the Technologies

4.3.3 Schedule Feasibility

- Gantt Chart

  Use a Gantt Chart, a scheduling chart, which marks off time periods in days, weeks or months. This chart must illustrate the different activities performed, when each begins, and when each terminates. The left-hand column are listed the tasks or activities to be performed. The right-hand portion of the chart has horizontal bars that show the duration of each activity.

  A Gantt chart is based upon time intervals and does not show the logical flow of information throughout a system. It shows the total time involved from the beginning of the project to its completion.

ACTIVITIES	6/19	6/26	7/3	7/10	7/17	…	12/18	12/25	1/8
Planning									
<subtask1>									
<subtask2>…									
System Analysis									
<subtask1>									
<subtask2>…									
System Design									
<subtask1>									
<subtask2>…									
System Development									
<subtask1>									
<subtask2>…									
System Implementation									
<subtask1>									
<subtask2>…									

4.3.4 Economic Feasibility

- Cost and Benefit Analysis
- Cost Recovery Scheme

4.3.5 Requirements Modeling

- Input
- Process
- Output
- Performance
- Control
- Either of the following two (2) or combined, whichever are applicable:
  - Data and Process Modeling
    - ✓ Context Diagram
    - ✓ Data Flow Diagram
    - ✓ System Flowchart
    - ✓ Program Flowchart (highlights only)
  - Object Modeling
    - ✓ Use Case Diagram
    - ✓ Class Diagram
    - ✓ Sequence Diagram
    - ✓ Activity Diagram

4.3.6 Risk Assessment/ Analysis

4.4 Design

4.4.1 Output and User-interface Design Forms

- Colors
- Typography
- Icons

4.4.2 Data Design

- Entity Relationship Diagram (preferably done in MS Access for presentation purposes [but MS Access is discouraged as DBMS])
- Data Dictionary

4.4.3 System Architecture

- Network Model
- Network Topology
- Security

4.5 Development

4.5.1 Software Specification

4.5.2 Hardware Specification

4.5.3 Program Specification

4.5.4 Programming Environment

- Front- End
- Back End
- Programming Considerations and Issues

4.5.5 Deployment Diagram

4.5.6 Test Plan

- Test Data

4.6 Verification, Validation, Testing

4.6.1 Unit Testing

4.6.2 Integration Testing

- Compatibility Testing
- Performance Testing
- Stress Testing
- Load Testing

4.6.3 System Testing

- System Function
- Test data
- Test Result and Observations

4.6.4 Acceptance Testing

4.7 Implementation Plan

This section describes everything about how the system is to interact with its environment. Included are the following kinds of items:

4.7.1 Physical Environment

- Where is the equipment to function?
- Is there one or several locations?
- How much physical space will be taken up by the system?

- Are they any environmental restrictions, such as temperature, humidity, or magnetic interface?
- What are the requirements of power, heating, or air conditioning?

4.7.2 Interfaces

- Is the input coming from one or more other systems?
- Is the output going one or more other system?
- Is there a prescribed way in which the data must be formulated?
- Is there a prescribed medium that the data must used

4.7.3 Functionally

- What will the system do?
- When the systems do it?
- How and when can the system be changed or enhanced?
- Are there constraints in execution speed, response time or throughout?

4.7.4 Data

- For both input and output, what should the format of data be?
- How often will it be received or sent?
- How accurate must it be?
- To what degree of precision must the calculations be made?
- How much data flows through the system?
- Must any data be retained for any period of time?

4.7.5 Security

- Must access to the system or to information to be controlled?
- How will ones user's data be isolated from others?
- How will one user programs be isolated from one other programs and from the OS?
- How often will the system be backed up?

4.8 Installation Processes

This selection should discuss what has been accomplished and activities from the existing systems to the new improved system. There are three basic approaches used in converting to a new system: an immediate / direct changeover a gradual step-by-step changeover, or a parallel system changeover, This section should discuss what type of installation or conversion approach will be used in the study and the different activities or process that will be performed to complete installation or conversion approach will be used in study and the different activities or process that will be performed to complete installation must be identified and discussed thoroughly.

5.0 Conclusion and Recommendations

Conclusions should discuss what has been accomplished in the study written in the objective to see clearly all significant aspects. It may subdivided into those that are primary aesthetic., Those that announce the results of an investigating and those that present a decision concerning a course of action . Also it may be numbered with respect to problems and sub- problems in study.

Recommendation should furnish future undertakings based on the analysis and conclusion of the study. It may also recommend potential applications of the study , other solutions, enhancement and/ or developments to the study.

REFERENCES

RESOURCE PERSONS

For each resource person:

<Full name and Title>	**Maria Jose**
<Profession>	Faculty
<Department>	College of Computer Studies
<Name of Institution>	Dominican College of Tarlac
<e-mail address/ tel. no>	mariajose@yahoo.com 045-322967

GLOSSARY

APPENDICES

Appendix A. Work Assignment
- identify the project involvement/ participation of each group members.

Appendix B. Definition of Terms

Appendix C. Evaluation Tool or Test Documents

Appendix D. User's Manual
- a user's guide in using the system. The manual is consist of sample screens and their corresponding description, instructions on how to perform specific tasks or use specific object, and a complete illustration of the system's usage (description that will assists the user on how to use the system).

Appendix E. Program Listing
- this contains a printed copy of all the programs, modules, functions and procedures of the developed system

Appendix F. Certifications

Include the following certifications:
- Certificate of Interview
- Certificate to Use Company's Data/ Information
- Certificate of Acceptance

Appendix G. Accomplished Forms

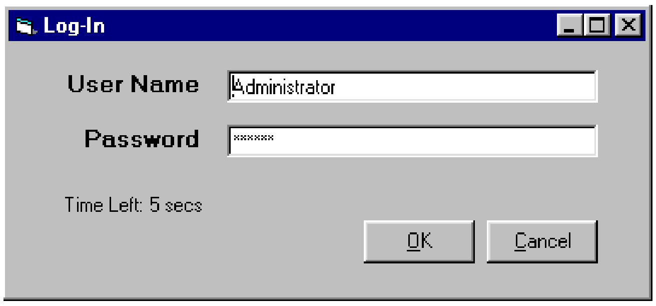
Appendix H Screen Design

Observe the following format:

Screen No. <screen#>
Screen Name: <name of the screen>
Narrative Overview: <brief description about the different components of the screen describing its functionality> Screen Layout: <include the screen layout/ design>

Example:

Screen No.1
Screen Name: Login Screen
Narrative Overview: Allows the user to try to enter the correct user name and password within twenty seconds.
Screen Layout:



- Reports

Appendix I. Others appendices may include the following:

- Transcript of Interview
  - contains documentation of all questions and answers obtained during the data gathering step.
- Survey Forms/Questionnaires
- Pictures showcasing the data gathering, investigation done (e.g. floor plan, layout building etc.)

Appendix J Curriculum Vitae (one page per member)

10.3 Documentation Standard Format

A. Paper
Size: 8.5x11
Orientation: Portrait (except for special diagram)
Substance: 20

B. Spacing: 1.5 inches
C. Indentation: 1 inch

D. Margins:
Top: 1 inch
Left: 1.5 inches
Bottom: 1 inch
Right: 1 inch
Gutter: 0
Header: 0.5
Footer: 0.5

E. Font
Sizes:
Heading 1: 12
Heading 2: 12
Heading 3 and Content: 11

Type: strictly Times New Roman
Color: Black (automatic)

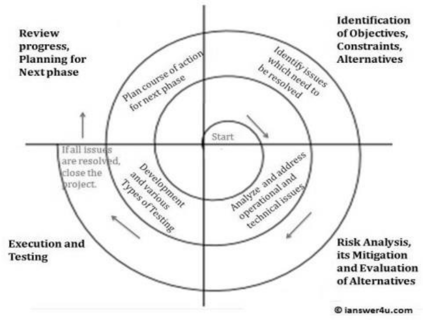
F. Pagination:
Bottom Right (no extra characters)
No page shown on first page of every chapter

G. Page Breaks
Page break is only used when starting a new chapter

H. Sample Layout for Tables

<Table No.>: <Table Title>

I. Sample Layout Figures



<Figure No.>: <Figure Title>

J. Sub-headings and text
All text and sub-headings should be in the following format as shown by an example below:

2.0 REVIEW OF RELATED STUDIES

2.1 Computer-Assisted Instruction

Computer-Assisted Instruction (CAI) refers to the use of computers to present drills, practice exercises, and tutorial sequences to the student.

One extensive type of CAI system is the PLATO [RALS1993]. It was developed by a group of engineers and educators in the Computer-based Education Laboratory at the University of Illinois, Urbane.

Stanford University operated CAI system to distribute instructional computing to a number of centers throughout the United States.

2.2 Intelligent Computer-Aided Instruction

Computer-Assisted Instruction (CAI) refers to the use of computers to present drills, practice exercises, and tutorial sequences to the student.

K. Bibliography (References and Citations)

This section deals with the nature of reference materials (e.g. Books, Unpublished Materials, Journals and Periodicals, etc.) if one wishes to read further in the area of problem or corollary areas. It also speaks of the researcher's awareness of the literature in his field and his critical resources.

Citations, as they appear within the text, should be coded to reflect the first four characters of the principal author's last name suffixed by the year of publication. Thus, [MILL1991] refers to the publication of Faron Miller for his work published 1991. If Miller has another publication for the same year, the code is appended with an alphabet in lowercase. Thus, a second publication of Miller should be coded as [MILL1991a], while a third publication should be coded as [MILL1991b].

Books:

<code> <author's name> (<year of publication>). <Book Title>, <site of publication>: <complete name of publisher>.

Example:

[CHIC1986] J M Chiclov (1986). An Introduction to Distributed and Parallel Computing. Hemel Hempstead: Prentice-Hall International (UK), Ltd.

Journal:

<code> <author's name> (<year of publication>). '<article title>', Journal Title, volume number (issue number), <pages where article could be found>.

Example:

[BAET1988] J C M Baeten & J A Bergstra (1988). 'Global Renaming Operators in Concrete Process Algebra', Information and Computation, 78(3), pp 205 – 245.

Conference:

<code> <author's name> (<year of publication>). '<article title>', In: Conference Name(editors of the proceedings, ed), <pages where article could be found>. <site of publication>: <complete name of publisher>.

Example:

[PARK1981] D H E Park (1981). 'Concurrency and Automata on Infinite Sequences', In: Fifth GI Conference (P Deussen, ed), pp 167 – 183. Berline: Springer-Verlag.

World Wide Web:

<code> <author of the page> (year). 'Homepage title'. URL site

Example:

[CRUZ1995] J Cruz (1996). 'The Home Page of Juan De La Cruz'. http://dlsu.edu.ph/aguinaldo.

***Note: Do not use the Traditional Style of Footnoting.

L. List of Appendices

Observe the following format:

APPENDICES

(in bold characters, font size 14)

Appendix <appendix letter> <Appendix Caption>

Appendix A. Work Assignment

DOMINICAN COLLEGE OF TARLAC
(In bold characters, font size 16)

<u><THESIS TITLE></u>
(In bold characters, underlined, font size 14)

A <Thesis/ Thesis Proposal> Presented to
Dominican College of Tarlac

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in Information Technology

by:

<last name, first name, middle initial of proponent 1>
<last name, first name, middle initial of proponent 2>
<last name, first name, middle initial of proponent 3>
<last name, first name, middle initial of proponent 4>

<Thesis Adviser's Name>
Thesis Adviser

<date of submission>
(month and year)

DOMINICAN COLLEGE OF TARLAC
(In bold characters, font size 16)

ADVISER'S RECOMMENDATION SHEET
(In bold characters, underlined, font size 14)

This <Capstone Project / Capstone Project Proposal> entitled

<Thesis Title>
(in bold characters, font size 14)

by:

<last name, first name, middle initial of proponent 1>
<last name, first name, middle initial of proponent 2>
<last name, first name, middle initial of proponent 3>
<last name, first name, middle initial of proponent 4>

And submitted in partial fulfillment of the requirements of the Bachelor of Science in <Program> degree has been examined and is recommended for acceptance and approval

<Thesis Adviser's Signature>
<Thesis Adviser's Name>
Thesis Adviser

<Date of submission>
Date

DOMINICAN COLLEGE OF TARLAC
(In bold characters, font size 16)

DEAN'S ACCEPTANCE SHEET
(In bold characters, underlined, font size 14)

This <Thesis/ Thesis Proposal> entitled

<**Thesis Title>**
(in bold characters, font size 14)

After having been recommended and approved is hereby accepted
by the <Name of department> Department
of Dominican College of Tarlac

< Dean's Signature >
<Dean's Name>
Dean

<Date of submission>
Date

DOMINICAN COLLEGE OF TARLAC
(In bold characters, font size 16)

PANEL'S APPROVAL SHEET
(In bold characters, underlined, font size 14)

This <Thesis/ Thesis Proposal> entitled

<Thesis Title>
(in bold characters, font size 14)

developed by:

<last name, first name, middle initial of proponent 1>
<last name, first name, middle initial of proponent 2>
<last name, first name, middle initial of proponent 3>
<last name, first name, middle initial of proponent 4>

after having been presented is hereby approved
by the following members of the panel

<Panelist 1's Signature>
Panelist

<Panelist 2's Signature>
Panelist

<Lead Panelist's Signature>
Lead Panelist

<date>

11.0 Areas of Research

A. Any System Development (Computerization) Work

1.1 Transaction Processing System
- Payroll System
- Sales and Inventory System
- Library System
- Accounting System
- Patient Information and Billing System (Hospital System)
- Enrollment System
- Others

1.2 Management Information System

1.3 Decision Support System

1.4 Geographic Information System

1.5 Scientific and Office Information System

Reminder: Information Systems may only be acceptable as a thesis project if there is an organization that will use the system.

B. Any Software Development Work

2.1 Multimedia System

2.2 Computer-Aided Instruction

2.3 Web Applications
- On-line Ordering System with Inventory Management
- On-line Hotel Reservation and Billing System
- On-line Job Application System
- On-line Registration System (Pre-Enrollment)
- Others

2.4 Artificial Intelligence
- Expert Systems
- Neural Networks
- Robotics
- Intelligent Agents
- ICAI

2.5 Software Engineering (SE)

2.6 Systems Analysis, Design and Implementation

2.7 Networking

Reminder: All software development works will be acceptable if the proponents can justify the need for such development. Thus, the software is expected to surpass all existing software of its kind.

C. Information Research Management

D. Unacceptable Thesis Projects

Transaction Processing Systems

Transaction Processing Systems (TPS) are computer-based versions of manual organization systems dedicated to handling the organizations' transactions.

General requirements:

a. complete file structure
   - normalized tables
   - use of master and transaction file
b. incorporation of back-up and recovery features (for projects which will use a DBMS that has no such feature)
c. archiving
d. complete data entry components
e. proper maintenance features
f. complete reports
g. data entry validation and error trapping
h. validation, security, data accuracy and integrity features
i. network set-up

Types of TPS:

1. Payroll System

A system designed to compute for the wage or salary due to each member in the organization
Important Considerations:

- number of employee – at least 50 employees
- types of employees (e.g. regular, contractual, part-time, consultant, etc.)
- time keeping system
- number of allowable absences/ leaves, deductions (due to late, absences, under time, cash advances, etc.)
- cut-off period; frequency of payroll period
- correctness and complexity of computations
- pay slip generation
- payroll summary report generation
- existence of table of variables (e.g. SSS table, Tax table, etc.)
- reports to other agencies (e.g. BIR, SSS, PAG-IBIG, etc.)
- valid test data – 50 employees

2. Sales and Inventory System

A system designed to monitor the quantity of inventory items (raw materials, production materials, and finished goods) to determine reorder or production point.
Important Considerations:

- number of inventory items – at least 1,000 items
- types of inventory items – 5 to 10 product lines
  5 to 10 items per product line
- warehouse capacity
- acquisition of items from supplier
- purchase order to supplier and purchase order of customers
- checking of critical levels (quantity ceiling and flooring), reorder point
- generation of reports (backorder list, reorder list, stock adjustment, etc.)
- frequency of reports (inventory summary)
- suppliers
- valid test data – 5 to 10 product lines
  5 to 10 items per product line

3. Library System

A system designed to perform the process involved in acquisition, cataloguing, and circulation of books.
Important Considerations:

- number of book titles – at least 2,000 books (CHED requirement)
- total number of books
- types of users
- number of students/ members – at least 500 students/ members
- borrowing system

- returning system
- reservation system
- cataloguing system
- computation of penalty
- CHED reports (for schools only)
- reports generated
- valid test data – 50 titles
  50 borrowers

4. Accounting System

A system designed to monitor the inflow and outflow of cash, goods, and services of an organization.

Important Considerations:

- accounts receivables monitoring
- accounts payable monitoring
- incorporation of aging technique
- posting of transaction on journal
- frequency of report generation
- valid test data – 30 accounts

5. Enrollment System

A system designed to perform the process involved in registration, advising, assessment and payment of students as well as scheduling of classes.

Important Considerations:

- number of students – at least 200 students
- number of sections – at least 2 sections per year level
- year levels (e.g. grades 1-6, 1^{st}-4^{th} year, etc.)
- registration
- assessment
- payments
- class schedule (scheduling/ sectioning)
- pre-requisites and co-requisites
- view tuition/fees billing statement
- courses and specializations (for Tertiary Level)
- student grades
- reports generated
- Student Information System (general information on student systems)
- Faculty Information System

6. Hotel Reservation and Billing System

A system designed to manage information about rooms, reservations, customers, and customer billing. The system provides functionality for making reservations, check in, and check out, in addition to generating reports and displays.

Important Considerations:

- number of rooms – at least 20 rooms
- number of customers – at least 100 customers
- customer information
- room information
- services and facilities
- check-in/ out procedures
- automatic billing system
- reservation and cancellation
- maintenance of fees
- reports generated
- valid test data – 50 customers

7. Patient Information and Billing System (Hospital System)

A system designed to perform process involved in admitting, diagnosing, medication, and billing of patients.

Important Considerations:

- number of beds – at least 20 beds
- total number of patients – at least 50 patients
- patient history
- medication
- diagnosis
- prognosis
- physicians' information
- data recorded and monitored for each patient
- automatic Billing System
- maintenance of fees (e.g. room rates, medicines, services, laboratory fee, etc.)
- valid test data – 50 patients

Management Information Systems

Management Information Systems (MIS) are computer-based systems designed to provide standard reports for managers about transaction data.
Important Considerations:

- types of reports generated
  - trend analysis report
  - summary report
  - exception report
  - on-demand report
  - ad-hoc report
  - frequency of reports
  - format of reports

Decision Support Systems

Decision Support Systems (DSS) are systems designed to help organization decision-makers make decisions.
Important Considerations:

- presence of three components
  - database
  - model base
  - dialogue module

Geographical Information Systems

Geographical Information System (GIS) is a computer system capable of assembling, storing, manipulating, and displaying geographically referenced information, i.e. data identified according to their locations. GIS technology can be used for scientific investigations, resource management, and development planning. For example, a GIS might allow emergency planners to easily calculate emergency response times in the event of a natural disaster, or a GIS might be used to find wetlands that need protection from pollution
Important Considerations:

- information retrieval
- topological modeling
- networks
- overlay (map layers)
- algorithms (e.g. shortest path algorithm, etc.)
- data output

Scientific and Office Information Systems

Scientific and Office Information Systems are systems used in the administration of organization or as a support to specific functions:

Important Considerations:

- quality of information
- quantity of information
- types of users
- functions to support
- availability of resources

Multimedia Systems

A software which encapsulates the representation of various digital media – text, images, audio, video, graphics, animation, music, etc.

Important Considerations:

- incorporation of at least 4 media (text, graphics, audio, video)
- incorporation of maintenance features; dynamic content management
- use of database(s) in storing information in various media
- nature of data, purpose, source and users of data
- search features, save and print functions
- organization/ category of topics or information
- users/ beneficiaries of the system
- user levels (e.g. systems administrator, viewers/ researchers, etc.)
- copyright of materials utilized for the system

Computer-Aided Instruction (CAI)

A software designed to facilitate learning. CAI utilizes multimedia integration to present lessons, give exams/ drills and provide necessary feedback.

Important Considerations:

- considerations for multimedia systems – incorporation of at least 4 media, incorporation of maintenance features, dynamic content management, etc.
- major categories – drills and practice, tutorials, simulation, games
- 2 to 3 methods of presenting the topics
- identification of the target age bracket/ users
- organization of topic/ discussions
- source/ basis of lessons/ topics
- level of difficulty of lessons and test items
- valid test items – source/ basis of questions
- random generation of test items
- analysis of results
- feedback of user performance
- testing – 50 items per topic
- required analysis/ design tools or methods – storyboarding, flowcharting, HIPO, others
- data flow diagrams must NOT be used in modeling CAI features and functions

Web Applications

Web Development focuses on using the latest technology to develop innovative and creative Web Applications.

General requirements:

a. considerations for multimedia systems
b. dynamic pages (components of a page must be stored and accessed in a database)
c. incorporation of maintenance features
d. use of database(s) in storing information in various media
e. nature of data, purpose, source and users of data
f. organization/ category of topics or information
g. standard conformance, portability, and clear navigation
h. search engine

i. report generation
j. meta tags & hyperlinks
k. e-consultation module – bulletin board or message board or send e-mail inquiry through browser
l. security
m. uploaded
n. required development tool/ programming languages: HTML, VBScripts, JavaScripts, XML, PHP, ASP, Perl and other web development programming languages

ACCEPTABLE PROJECTS (HOST COMPANY IS NECESSARY):

1. **On-line Ordering System with Inventory Management**
   Important considerations:
   - considerations for (TPS) inventory system – number of inventory items, types of inventory items, etc.
   - company profile, policies, terms and conditions, location, contact details
   - customer profile
   - online product catalog (product information such as price and description, etc.)
   - monitoring of inventory availability
   - ordering module
   - online approval/disapproval of transactions
   - cancellation of orders/transactions
   - automatic and prompt feedback mechanism (for product inquiry, ordering, monitoring, tracking, etc.)
   - delivery and payment
   - FAQ (frequently asked questions module)
   - Dynamic Advertisement module
   - Product Search Engine module

2. **On-line Hotel Reservation and Billing System**
   Important considerations:
   - considerations for (TPS) hotel reservation and billing system – no. of rooms, customers, etc.
   - company profile, policies, terms and conditions, location, contact details
   - provision of interactive location maps, vicinity maps, and on-line sample model unit
   - detailed listing and information of facilities, amenities, services, terms of payment, packages
   - on-line approval/ disapproval/ cancellation of transactions (reservation)
   - customer profile
   - FAQ (frequently asked questions module)
   - Dynamic Advertisement module – What's new in our product, Offers and Promotions
   - Product Search Engine module

3. **On-line Job Application System**
   Important considerations:
   - number of applicants – at least 100 applicants
   - number of job offerings – at least 10
   - types of job
   - host company, client/ employers and applicants profile
   - individual accounts for administrator, client/ employers and applicant
   - job orders from clients/employers
   - job categories, descriptions and requirements
   - posting of job vacancies, descriptions and qualifications
   - job search
   - online application feature
   - job fit assessment feature – assessment of personality and qualification
   - applicant manager feature
     - applicant tracker – allows you to view applicants who have recently completed one of the steps in the hiring process
     - open positions – view all applicants who have applied for a specific open position by selecting the position title
     - applicant search – locate an individual applicant by searching with specific applicant criteria
     - add new applicant – manually add applicants who did not complete the Online Application
   - "job matching" facility

- module/function for trimming down of applicants based on employer's/client's specifications and requirements
- automatic and prompt feedback mechanism (for job inquiry, application, monitoring, tracking, etc.)

4. On-line Registration System (Pre-Enrollment)

Important Considerations:

- applicable for schools offering tertiary level only
- considerations for (TPS) enrollment system – no. of students, section, year level, etc.
- online registration
- online assessment
- curriculum tracking and update
- class schedule
- pre-requisites and co-requisites
- view tuition
- programs, courses and specializations (for Tertiary Level)
- student grades
- reports generated
- Student Information System (general information on student systems)
- Faculty Information System
- School Information
    - mission and vision of the school
    - history of the school
    - organizational structure of the school
    - current heads and authorities
    - student services (registrar, faculty, classrooms, laboratories, library, career office, guidance, etc.)
    - school calendar, student events, activities, projects and programs
    - academic programs/ courses offered and career opportunities per program
    - general table of tuition fees per course per year

Additional Reference

http://teachers.net/manual/
http://harry.cciflorida.com/HTML3/home/work/opini/www/websiteplan.html
http://www.phpbuilder.com/columns/angus20010304.php3
http://www.w3.org/TR/WAI-WEBCONTENT/

Artificial Intelligence (AI)

Types of AI:

1. ***Expert Systems (ES)*** – a computer-based system designed to mimic the performance of human experts.
   Important Considerations:
    - database
    - dialogue module
    - inference engine
    - knowledge acquisition technique
    - expert system programming language
2. ***Neural Networks*** – a system modeled after the neurons (nerve cells) in a biological nervous system and intended to stimulate the way in which our brain processes information, learns and remember.
3. ***Robotics*** – the integration of computers and industrial robots, and is more often than not associated with unmanned assembly lines.
4. ***Intelligent Agent***s
5. ***ICAI***

Software Engineering

Software Engineering is the establishment and use of sound engineering principles in order to obtain economical software that is reliable and works efficiently on real machines.

Focus of Research:

- Paradigm of software engineering based on new approaches in programming
- Cost models
- Systems Software – a type of software that tells the CPU what to do
  Example: Operating System, Compilers, Interpreters, Utilities, etc.
- Other Software Packages
  Example: Desktop Publishing, Graphics Editor, Word Processor, Spreadsheets, CAD, Communications Software, etc.

Systems Analysis and Design

Systems Analysis and Design and Implementation is a complex organizational process whereby computer-based information systems are developed and maintained.
Focus of Research: Automated Tool that combines both process and data views of systems

Networking

Networking is the process of connecting computer devices and circuits for transferring data from one computer to another.
Focus of Research: Network topology implementation

Information Research Management

Management of information which evolves on the concept that information is a major corporate resource that should be managed in the same manner as other assets such as land, people, etc., are managed.

- Information Systems Plan for an organization

Unacceptable Thesis Projects

The following are the unacceptable thesis projects:

- DAMATH
- Video Rental System
- Games – card games, non-educational games
- Record Keeping Systems
- Monitoring System
- Web Sites – Barangay, Municipality, City, Provincial, etc.

Please note that unacceptable projects are not limited to the above-mentioned projects. DCT could still reject/ disapprove projects depending on its scope, feasibility, practicality and originality.

CAPSTONE PROJECT INTELLECTUAL PROPERTY AND OWNERSHIP POLICY GUIDELINES

1. Ownership and Transfer of Intellectual Property

All capstone projects developed by students as part of their academic requirements shall initially be recognized as the intellectual property of the student developers. Upon completion, submission, and approval of the capstone project, ownership of the intellectual property will be transferred to the College of Computer Studies (CCS), subject to institutional policies and written acknowledgment by the student developers.

Such transfer includes, but is not limited to, rights over:

- Source codes and system architecture
- Project documentation and technical reports
- Multimedia assets and design elements
- Deployment, enhancement, maintenance, and institutional use of the system

The transfer of intellectual property shall take effect upon:

- Final approval of the capstone project, and
- Signing of the Intellectual Property Transfer and Acknowledgment Form by the student developers.

2. School's Rights and Usage

While ownership remains with the students, the College / University reserves the non-exclusive, royalty-free right to:

- Archive the capstone project for academic records
- Display or demonstrate the project for instructional, accreditation, research, and promotional purposes
- Use screenshots, descriptions, or non-sensitive portions of the project in official publications, exhibits, or events

Proper acknowledgment of the student developers shall always be observed.

3. Adviser and Panel Contributions

Advisers, panel members, and faculty members who provide substantial intellectual, technical, or creative contributions to the development of a capstone project shall be recognized as co-authors of the work. Any contribution that goes beyond academic supervision, consultation, or evaluation automatically constitutes co-authorship and shall be duly acknowledged in all official submissions, presentations, publications, and related outputs of the capstone project.

In addition, advisers and faculty contributors are granted the right to use the capstone project and its outputs for legitimate academic purposes, including but not limited to instruction, research, publication, presentation, accreditation, training, and professional development, provided that proper attribution to the student developer(s) and the institution is observed. Such academic use shall be non-commercial in nature unless otherwise covered by a separate written agreement approved by the DCT-CCS Department.

4. Sponsored or Externally Funded Projects

For capstone projects developed in collaboration with:

- Industry partners
- Government agencies
- External organizations

Ownership and usage rights shall be governed by a separate Memorandum of Agreement (MOA). Students must be informed in writing of any modified ownership conditions prior to project approval.

5. Use of Third-Party Materials

Students are responsible for ensuring that:

- All third-party libraries, APIs, datasets, images, and media are properly licensed
- Open-source components are used in compliance with their respective licenses
- Proper citations and attributions are included in the documentation

Any violation of intellectual property laws shall be the responsibility of the student developers.

6. Commercialization and Future Development

Students may choose to:

- Further develop, publish, or commercialize their capstone project after completion
- Register copyrights, patents, or trademarks in their names

The College shall not claim financial rights or ownership unless explicitly agreed upon in writing.

7. Confidentiality and Data Protection

Capstone projects that involve sensitive, proprietary, or personal data must comply with:

- Data Privacy Act of 2012 (RA 10173)
- Institutional data protection policies

Confidential information must not be publicly disclosed without proper authorization.

8. Acknowledgment Requirement

Any public presentation, deployment, or publication of the capstone project shall acknowledge:

- The institution
- The DCT-College of Computer Studies
- The academic program under which the project was developed

9. Policy Acceptance

Submission of a capstone proposal and final output constitutes the student's acknowledgment and acceptance of this Intellectual Property and Ownership Policy.

10. Policy Acknowledgment and Acceptance

By submitting the capstone proposal, final manuscript, and project output, the student researcher(s) **formally acknowledge, understand, and agree** to the provisions of this Intellectual Property and Co-Authorship Policy. This acknowledgment signifies acceptance of the roles, rights, and responsibilities of all contributors, including mandatory co-authorship where applicable, and confirms compliance with institutional and academic standards.

ACADEMIC INTEGRITY, PLAGIARISM, AND RELATED VIOLATIONS GUIDELINES FOR CAPSTONE PROJECTS

1. Policy Statement

The College of Computer Studies upholds the principles of academic integrity, honesty, originality, and ethical scholarship. All capstone papers, system outputs, documentation, and related submissions must be the original work of the student researchers and must comply with institutional and CHED standards.

Any form of plagiarism, falsification, misrepresentation, or other academic anomalies shall be treated as a serious offense and shall be subject to appropriate disciplinary action.

2. Definition of Plagiarism

Plagiarism includes, but is not limited to:

- Copying text, ideas, code, designs, or data from published or unpublished sources without proper citation
- Paraphrasing substantial portions of another work without acknowledgment
- Submitting work purchased, borrowed, or generated by another person or entity and claiming it as one's own
- Reusing one's own previously submitted academic work without proper disclosure (self-plagiarism)
- Plagiarism applies to written manuscripts, source code, system designs, multimedia assets, and research outputs.

3. Other Academic Anomalies

Academic anomalies include, but are not limited to:

- Fabrication or falsification of data, results, or system features
- Misrepresentation of system functionality or project scope
- Unauthorized use of artificial intelligence or automated tools in violation of course or institutional rules
- Submission of incomplete, manipulated, or tampered documents
- Improper authorship attribution or omission of contributors

4. Plagiarism Detection and Review

All capstone papers and related documents shall be subjected to plagiarism detection and academic integrity review through the following mechanisms:

- **Mandatory Plagiarism Checking Prior to Oral Defense**
  All capstone papers must undergo plagiarism checking through the DCT- Research Office or the DCT Library prior to the scheduled oral defense. Only papers that have been officially cleared by the Research Office or Library shall be allowed to proceed to oral defense.
- **Plagiarism Checking After Final Defense and Prior to Publication**
  Upon successful completion and acceptance of the capstone project during the final oral

defense, the revised and approved final manuscript shall again be subjected to plagiarism checking through the Research Office or Library before the capstone book, bound manuscript, or institutional publication is released or archived.

- **Manual Review**
  In addition to software-based plagiarism detection, capstone papers may undergo manual review by advisers, panel members, or authorized faculty to assess proper citation, originality, and ethical compliance.
- **Scope of Review**
  The College reserves the right to conduct plagiarism checks at any stage of the capstone process, including but not limited to proposal defense, midterm evaluation, final defense, and post-submission, should concerns arise.

5. Acceptable Similarity Index

The acceptable similarity index shall be determined by the Research Office. Any submission that exceeds the prescribed threshold or shows evidence of improper attribution shall be subject to further review, regardless of the percentage score.

A low similarity index does not automatically exempt a paper from investigation if intentional plagiarism or anomalies are evident.

6. Sanctions and Disciplinary Actions

If plagiarism or other academic anomalies are confirmed, one or more of the following sanctions may be imposed, depending on the severity and frequency of the offense:

- Mandatory revision and resubmission of the capstone paper
- Re-defense of the capstone project
- Automatic failure of the capstone subject
- Delay or disqualification from graduation
- Referral to the DCT Disciplinary or Ethics Committee
- Other penalties in accordance with DCT and CHED policies

7. Due Process

Students shall be afforded due process, including:

- Written notification of the alleged violation
- Opportunity to explain or respond to the findings
- Review and deliberation by the appropriate academic body
- Final decisions shall be based on documented evidence and institutional rules.

8. Adviser and Panel Responsibility

Advisers and panel members are responsible for:

- Guiding students on proper citation, ethical research, and originality
- Reporting suspected cases of plagiarism or anomalies
- Ensuring compliance with academic integrity standards
- Failure to disclose known violations may result in further institutional review.

9. Policy Acknowledgment

Submission of a capstone proposal, manuscript, system output, or defense requirements constitutes the student researchers’ acknowledgment and acceptance of this Academic Integrity and Plagiarism Policy.

Ignorance of this policy shall not be accepted as a valid defense.

10. Effectivity

This policy shall take effect upon approval by the College of Computer Studies and shall remain enforceable throughout the capstone duration and after project completion, if violations are discovered subsequently.

ACM Word Template for SIG Site

1st Author	2nd Author	3rd Author
1st author's affiliation	2nd author's affiliation	3rd author's affiliation
1st line of address	1st line of address	1st line of address
2nd line of address	2nd line of address	2nd line of address
Telephone number, incl. country code	Telephone number, incl. country code	Telephone number, incl. country code
1st author's email address	2nd E-mail	3rd E-mail

ABSTRACT

In this paper, we describe the formatting guidelines for ACM SIG Proceedings.

Categories and Subject Descriptors
D.3.3 [**Programming Languages**]: Language Constructs and Features – *abstract data types, polymorphism, and control structures*. This is just an example, please use the correct category and subject descriptors for your submission. The ACM Computing Classification Scheme: http://www.acm.org/class/1998/

General Terms
Your general terms must be any of the following 16 designated terms: Algorithms, Management, Measurement, Documentation, Performance, Design, Economics, Reliability, Experimentation, Security, Human Factors, Standardization, Languages, Theory, Legal Aspects, and Verification.

Keywords
Keywords are your own designated keywords.

INTRODUCTION

The proceedings are the records of the conference. ACM hopes to give these conference by-products a single, high-quality appearance. To do this, we ask that authors follow some simple guidelines. In essence, we ask you to make your paper look exactly like this document. The easiest way to do this is simply to download a template from [2], and replace the content with your own material.

PAGE SIZE

All material on each page should fit within a rectangle of 18 x 23.5 cm (7" x 9.25"), centered on the page, beginning 2.54 cm (1") from the top of the page and ending with 2.54 cm (1") from the bottom. The right and left margins should be 1.9 cm (.75"). The text should be in two 8.45 cm (3.33") columns with a .83 cm (.33") gutter.

TYPESET TEXT

Normal or Body Text

Please use a 9-point Times Roman font, or other Roman font with serifs, as close as possible in appearance to Times Roman in which these guidelines have been set. The goal is to have a 9-point text, as you see here. Please use sans serif or non-proportional fonts only for special purposes, such as distinguishing source code text. If Times Roman is not available, try the font named Computer Modern Roman. On a Macintosh, use the font named Times. Right margins should be justified, not ragged.

Title and Authors

The title (Helvetica 18-point bold), authors' names (Helvetica 12-point) and affiliations (Helvetica 10-point) run across the full width of the page – one column wide. We also recommend phone number (Helvetica 10-point) and e-mail address (Helvetica 12-point). See the top of this page for three addresses. If only one address is needed, center all address text. For two addresses, use two centered tabs, and so on. For more than three authors, you may have to improvise.[1]

First Page Copyright Notice

[1] If necessary, you may place some address information in a footnote or in a named section at the end of your paper.

Please leave 3.81 cm (1.5") of blank text box at the bottom of the left column of the first page for the copyright notice.

Subsequent Pages

For pages other than the first page, start at the top of the page, and continue in double-column format. The two columns on the last page should be as close to equal length as possible.

Table 1. Table captions should be placed above the table

Graphics	Top	In-between	Bottom
Tables	End	Last	First
Figures	Good	Similar	Very well

References and Citations

Footnotes should be Times New Roman 9-point, and justified to the full width of the column.

Use the "ACM Reference format" for references – that is, a numbered list at the end of the article, ordered alphabetically and formatted accordingly. See examples of some typical reference types, in the new "ACM Reference format", at the end of this document. Within this template, use the style named *references* for the text. Acceptable abbreviations, for journal names, can be found here: http://library.caltech.edu/reference/abbreviations/

The references are also in 9 pt., but that section (see Section 7) is ragged right. References should be published materials accessible to the public. Internal technical reports may be cited only if they are easily accessible (i.e. you can give the address to obtain the report within your citation) and may be obtained by any reader. Proprietary information may not be cited. Private communications should be acknowledged, not referenced (e.g., "[Robertson, personal communication]").

Page Numbering, Headers and Footers

Do not include headers, footers or page numbers in your submission. These will be added when the publications are assembled.

FIGURES/CAPTIONS

Place Tables/Figures/Images in text as close to the reference as possible (see Figure 1). It may extend across both columns to a maximum width of 17.78 cm (7").

Captions should be Times New Roman 9-point bold. They should be numbered (e.g., "Table 1" or "Figure 2"), please note that the word for Table and Figure are spelled out. Figure's captions should be centered beneath the image or picture, and Table captions should be centered above the table body.

SECTIONS

The heading of a section should be in Times New Roman 12-point bold in all-capitals flush left with an additional 6-points of white space above the section head. Sections and subsequent sub- sections should be numbered and flush left. For a section head and a subsection head together (such as Section 3 and subsection 3.1), use no additional space above the subsection head.

Subsections

The heading of subsections should be in Times New Roman 12-point bold with only the initial letters capitalized. (Note: For subsections and sub subsections, a word like *the* or *a* is not capitalized unless it is the first word of the header.)

Sub subsections

The heading for sub subsections should be in Times New Roman 11-point italic with initial letters capitalized and 6-points of white space above the sub subsection head.

Sub subsections

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The heading for sub subsections should be in Times New Roman 11-point italic with initial letters capitalized.

ACKNOWLEDGMENTS

Our thanks to ACM SIGCHI for allowing us to modify templates they had developed.

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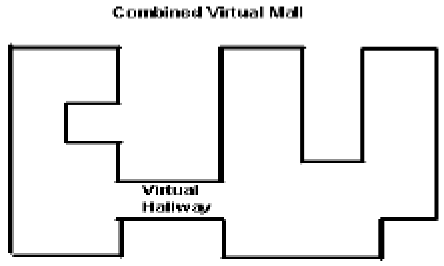

Figure 1. Insert caption to place caption below figure.

[2] Ding, W. and Marchionini, G. 1997 A Study on Video Browsing Strategies. Technical Report. University of Maryland at College Park.

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Columns on Last Page Should Be Made As Close As Possible to Equal Length